

Question 1 | Correct | Mark 1.00 out of 1.00**Problem Statement:**

Given a sorted array and a value x, the floor of x is the largest element in array smaller than or equal to x. Write divide and conquer algorithm to find floor of x.

Input Format

First Line Contains Integer n – Size of array

Next n lines Contains n numbers – Elements of an array

Last Line Contains Integer x – Value for x

Output Format

First Line Contains Integer – Floor value for x

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main(){
3      int n,i,x,ans=-1,mid,high,low;
4      scanf("%d",&n);
5      int a[n];
6      for(i=0;i<n;i++){
7          scanf("%d",&a[i]);
8      }
9      scanf("%d",&x);
10     low=0;
11     high=n-1;
12     while(low<=high){
13         mid=(low+high)/2;
14         if(a[mid]<=x){
15             low=mid+1;
16             ans=a[mid];
17         }
18         else
19             high=mid-1;
```

```
20     }  
21     printf("%d",ans);  
22  
23 }
```

	Input	Expected	Got	
✓	6 1 2 8 10 12 19 5	2	2	✓
✓	5 10 22 85 108 129 100	85	85	✓